



ABSTRACT

Solid waste management has become one of the most significant environmental and public health challenges associated with rapid urbanization, population growth, and changing lifestyles. Improper disposal of solid waste leads to environmental pollution, the spread of infectious diseases, degradation of natural resources, and adverse effects on urban biodiversity. The present study, entitled "**Solid Waste Management**," aims to examine the existing waste management practices, public awareness, and the environmental and health impacts of improper waste disposal, with special reference to urban areas.

The study focuses on understanding the sources, types, collection, segregation, transportation, treatment, and disposal of solid waste while evaluating public perception and participation in sustainable waste management practices. Both primary and secondary data were utilized through field observations, questionnaires, interviews, and a review of relevant literature. The findings indicate that inadequate waste segregation, irregular collection services, limited public awareness, and improper disposal methods contribute significantly to environmental degradation and public health risks. The study also highlights the impact of unmanaged waste on urban wildlife, disease vectors, and ecosystem balance, emphasizing its relevance to the field of Zoology.

The report concludes that effective solid waste management requires an integrated approach involving government authorities, local communities, educational institutions, and individuals. Promoting waste segregation at source, recycling, composting, public awareness campaigns, and environmentally sustainable waste management policies can significantly reduce pollution and improve public health. The recommendations presented in this study aim to encourage responsible waste management practices and contribute to environmental conservation and sustainable urban development.

Keywords: Solid Waste Management, Urban Environment, Waste Segregation, Recycling, Public Health, Environmental Pollution, Urban Biodiversity, Sustainable Development.